Employee Data Management System

Write a querv to display departments name, and the number of emplovees in each department

```sql
SELECT
    DEPARTMENT_NAME AS Department,
    COUNT(employees.EMPLOYEE_ID) AS Number_Of_Employees
FROM
    departments,
    employees
WHERE
    departments.DEPARTMENT_ID = employees.DEPARTMENT_ID
GROUP BY
    DEPARTMENT_NAME;
```

Department	Number_Of_Employees
Accounting	2
Administration	1
Executive	3
Finance	6
Human Resources	1
IT	5
Marketing	2
Public Relations	1
Purchasing	6
Sales	34
Shipping	45

2) **Write a querv to display the department name, and average salarv of emplovees in each department**

```sql
SELECT
    DEPARTMENT_NAME AS Department,
    AVG(employees.SALARY) AS Average_Salary_Of_Employees
FROM
    departments,
    employees
WHERE
    departments.DEPARTMENT_ID = employees.DEPARTMENT_ID
GROUP BY
    DEPARTMENT_NAME;
```

Department	Average_Salary_Of_Employees
Accounting	10150.000000
Administration	4400.000000
Executive	19333.333333
Finance	8600.000000
Human Resources	6500.000000
IT	5760.000000
Marketing	9500.000000
Public Relations	10000.000000
Purchasing	4150.000000
Sales	8955.882353
Shipping	3475.555556

3) **Write a query to display the names or employees having salary greater than average or their department**

```
select Q1.Full_Name from (select (departments.DEPARTMENT_NAME)DP1,CONCAT(employees.FIRST_NAME," ",employees.LAST_NAME)Full_Name,
(employees.SALARY)S1 from employees,departments where employees.DEPARTMENT_ID = departments.DEPARTMENT_ID)Q1,(select
avg(employees.SALARY)avgsalary,(departments.DEPARTMENT_NAME)DP2 from employees,departments where employees.DEPARTMENT_ID =
departments.DEPARTMENT_ID group by departments.DEPARTMENT_NAME)Q2 where Q1.S1 > Q2.avgsalary && Q1.DP1 = Q2.DP2 group by Q1.Full_Name order by
q1.dp1;
```

Full_Name
Shelley Higgins
Steven King
Daniel Faviet
Nancy Greenberg
Alexander Hunold
Bruce Ernst
Michael Hartstein
Den Raphaely
John Russell
Gerald Cambrault
David Bernstein
Patrick Sully
Danielle Greene
Tayler Fox
Karen Partners
Eleni Zlotkey
Peter Hall
Allan McEwen
Lisa Ozer
Ellen Abel
Alberto Errazuriz
Peter Tucker
Janette King
Clara Vishney
Harrison Bloom

Full_Name
Jennifer Dilly
Adam Fripp
Kevin Mourgos
Alexis Bull
Sarah Bell
Payam Kaufling
Renske Ladwig
Kelly Chung
Britney Everett
Matthew Weiss
Shanta Vollman
Trenna Rajs
Nandita Sarchand

4) **Write a query to display the region having the highest employee salary**

```
SELECT
    regions.REGION_NAME AS Region
FROM
    regions,
    countries,
    departments,
    employees,
    locations
WHERE
    regions.REGION_ID = countries.REGION_ID && countries.COUNTRY_ID = locations.COUNTRY_ID && locations.LOCATION_ID = departments.LOCATION_ID && departments.DEPARTMENT_ID = employees.DEPARTMENT_ID && employees.SALARY =(
    SELECT
        MAX(employees.SALARY)
    FROM
        employees
)
```

←T→				Region
☐	Edit	Copy	Delete	Americas

5) **Write a query to display the names of employees who changed their jobs**

```
select DISTINCT CONCAT(employees.FIRST_NAME," ",employees.LAST_NAME)Full_Name from employees,job_history where
employees.EMPLOYEE_ID = job_history.EMPLOYEE_ID
```

Full_Name
Neena Kochhar
Lex De Haan
Den Raphaely
Payam Kaufling
Jonathon Taylor
Jennifer Whalen
Michael Hartstein

6) **Write a query to display the name of countries, count of employees who changed their jobs**

```
select (countries.COUNTRY_NAME)Country, count(DISTINCT employees.EMPLOYEE_ID)Number_Of_Employees from
employees,job_history,departments,locations,countries where job_history.EMPLOYEE_ID = employees.EMPLOYEE_ID &&
departments.DEPARTMENT_ID = job_history.DEPARTMENT_ID && departments.LOCATION_ID = locations.LOCATION_ID &&
locations.COUNTRY_ID = countries.COUNTRY_ID  && employees.EMPLOYEE_ID = job_history.EMPLOYEE_ID group by
countries.COUNTRY_NAME
```

Country	Number_Of_Employees
Canada	1
United Kingdom	1
United States of America	5

7) **Write a query to display the month the count of managers in each department**

```sql
SELECT
    (departments.DEPARTMENT_NAME) Department,
    MONTH(employees.HIRE_DATE) MONTH,
    COUNT(DISTINCT employees.MANAGER_ID) Number_Of_Managers
FROM
    employees,
    departments
WHERE
    employees.MANAGER_ID != 0 AND employees.DEPARTMENT_ID = departments.DEPARTMENT_ID
GROUP BY
    departments.DEPARTMENT_NAME
```

Department	Month	Number_Of_Managers
Accounting	10	2
Administration	9	1
Executive	6	1
Finance	6	2
Human Resources	9	1
IT	6	2
Marketing	9	2
Public Relations	9	1
Purchasing	7	2
Sales	8	6
Shipping	7	6

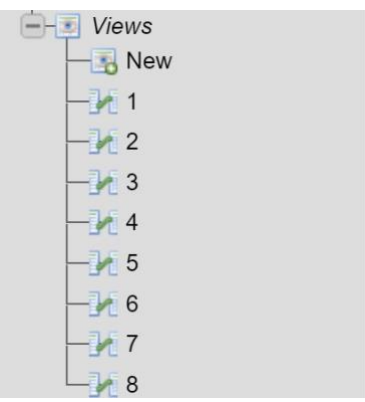
8) **Write a query to display the name or managers, count of employees under him**

```
select Q2.Full_Name,Q1.Number_Of_Employees from (select DISTINCT
(employees.MANAGER_ID)MANAGER_ID,count(employees.MANAGER_ID)Number_Of_Employees from employees where employees.MANAGER_ID !=
"0" group by employees.MANAGER_ID)Q1,(select (employees.EMPLOYEE_ID)e1,CONCAT(employees.FIRST_NAME,"
",employees.LAST_NAME)Full_Name from employees)Q2 where Q2.e1 = Q1.MANAGER_ID;
```

Full_Name	Number_Of_Employees
Steven King	14
Neena Kochhar	5
Lex De Haan	1
Alexander Hunold	4
Nancy Greenberg	5
Den Raphaely	5
Matthew Weiss	8
Adam Fripp	8
Payam Kaufling	8
Shanta Vollman	8
Kevin Mourgos	8
John Russell	6
Karen Partners	6
Alberto Errazuriz	6
Gerald Cambrault	6
Eleni Zlotkey	6
Michael Hartstein	1
Shelley Higgins	1

9) **After finishing the queries, make a view for each**

	Table	Action						Rows	Type	Collation	Size	Overhead
☐	1	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	2	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	3	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	4	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	5	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	6	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	7	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
☐	8	Browse	Structure	Search	Insert	Edit	Drop	~0	View	---	-	-
	8 tables	Sum						~0	InnoDB	utf8mb4_general_ci	0 B	0 B



10) **Explain the following; how to make a view updatable? What is local and cascade? Processing algorithms?**

- A view is updatable when it's SELECT statement must not contain any aggregate functions(MIN,MAX,AVG,SUM,COUNT),DISTINCT,GROUP BY,HAVING,UNION,UNION ALL,LEFT JOIN,OUTER JOIN,Subquery in SELECT or WHERE that refers to a table from

FROM, reference to a non-updatable view, reference only to literal values, multiple references to any column of the base table.

- When a user creates a view based on other views MYSQL checks the rules in the dependent views for data consistency using WITH CHECK OPTION.To determine the scope of the check MYSQL provides two options: LOCAL and CASCADE.
    1. WITH CASCADED CHECK OPTION:

```
CREATE TABLE t1 (
    c INT
);
```

We create a table t1 and a integer variable c

```
CREATE OR REPLACE VIEW v1
AS
    SELECT
        c
    FROM
        t1
    WHERE
        c > 10;
```

Then we create a view v1 based on t1 with a condition that c is greater then 10

```
INSERT INTO v1(c)
VALUES (5);
```

We then inserted 5 to c which is not greater then 10
The statement worked because we did not specify any WITH CHECK OPTION.

```sql
CREATE OR REPLACE VIEW v2
AS
    SELECT c FROM v1
WITH CASCADED CHECK OPTION;
```

Next we created a view v2 based on v1 with cascaded check option

```sql
INSERT INTO v2(c)
VALUES (5);
```

We inserted 5 to c

```
Error Code: 1369. CHECK OPTION failed 'classicmodels.v2'
```

We received an error because it has a WITH CHECK OPTION v2 is dependent on v1 which has a condition of c>10.

```sql
CREATE OR REPLACE VIEW v3
AS
    SELECT c
    FROM v2
    WHERE c < 20;
```

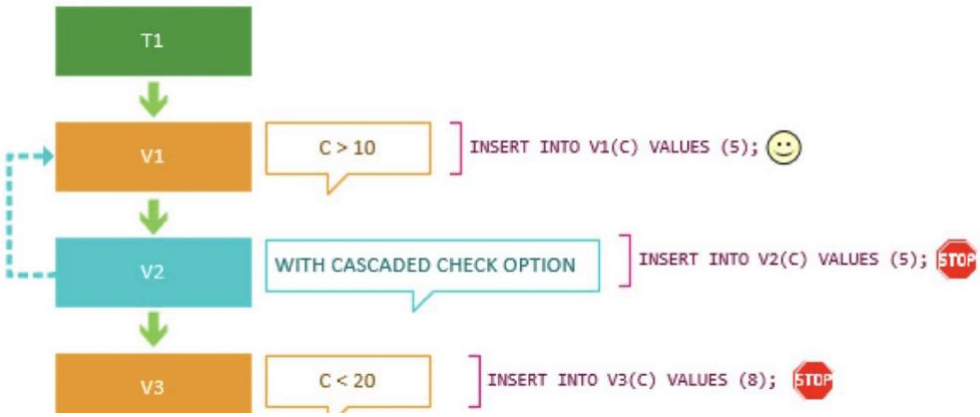
We create a new view v3 from v2 with c < 20

```sql
INSERT INTO v3(c)
VALUES (8);
```

We insert 8 to c

```
Error Code: 1369. CHECK OPTION failed 'classicmodels.v3'
```

We received an error because v3 is dependent on v2 and v2 has a WITH CHECK OPTION and v2 is dependent on v1 which condition states 8 < 10 and v3 does not have a WITH CHECK OPTION.
To conclude a view that uses a WITH CASCADED CHECK OPTION checks the rules of the view and also the rules of the underlying views recursively.
It basically checks the rules of all dependent views.



2. WITH LOCAL CHECK OPTION

```
ALTER VIEW v2 AS
    SELECT
        c
    FROM
        v1
WITH LOCAL CHECK OPTION;
```

```
INSERT INTO v2(c)
VALUES (5);
```

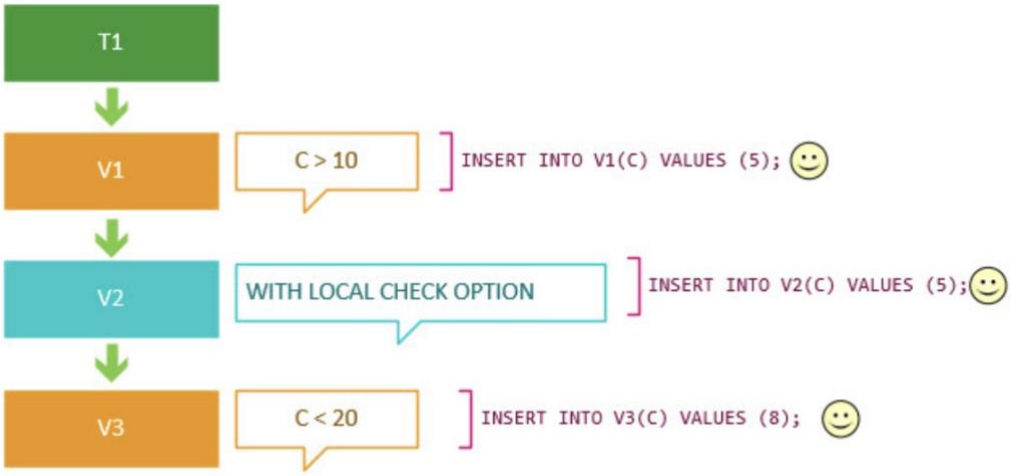
Modifying v2 to WITH LOCAL CHECK OPTION and inserting 5 to c

The statement works because v2 doesn't have any rules and v2 is dependent on v1 and v1 doesn't have any WITH CHECK OPTION so it skips checking the rules in v1.

```
INSERT INTO v3(c) VALUES (8);
```

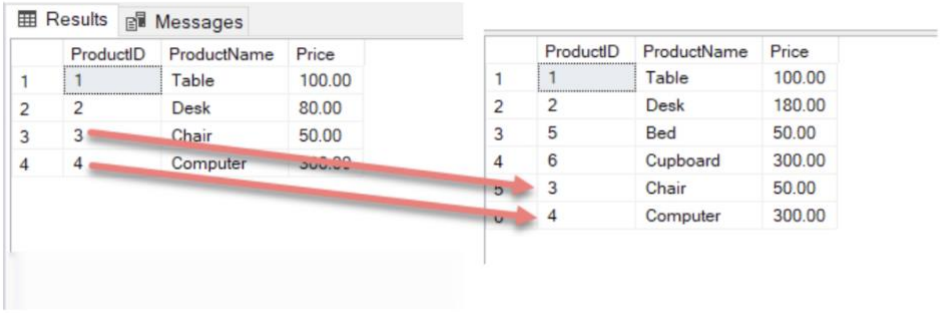
It succeeds in this case because it did not check the rules of v1 view because of the WITH LOCAL CHECK OPTION of v2.
To conclude a view that uses a WITH LOCAL CHECK OPTION checks the rules of views that have a WITH LOCAL CHECK OPTION and a WITH CASCADED CHECK OPTION.

- The create or alter a view statements have an optional clause: Algorithm.It determines



how MYSQL process a view and has one of three values: Merge,Temptable and Undefined..

1. Merge: the text of a statement that refers to the view and the view definition are merged such that parts of the view definition replace corresponding parts of the statement.
   In this example we merged the last two parts of t1 into t2.



2. Temptable: The results from the view are retrieved into a temporary table which then is used to execute the statement.
   In this example we created a temporary table top_customers and selected customerNumber,customerName and sales.

```
CREATE TEMPORARY TABLE top_customers
SELECT p.customerNumber,
       c.customerName,
       ROUND(SUM(p.amount),2) sales
FROM payments p
INNER JOIN customers c ON c.customerNumber =
GROUP BY p.customerNumber
ORDER BY sales DESC
LIMIT 10;
```

```
SELECT
    customerNumber,
    customerName,
    sales
FROM
    top_customers
ORDER BY sales;
```

customerNumber	customerName	sales
146	Saveley & Henriot, Co.	130305.35
321	Corporate Gift Ideas Co.	132340.78
276	Anna's Decorations, Ltd	137034.22
187	AV Stores, Co.	148410.09
323	Down Under Souveniers, Inc	154622.08
148	Dragon Souveniers, Ltd.	156251.03
151	Muscle Machine Inc	177913.95
114	Australian Collectors, Co.	180585.07
124	Mini Gifts Distributors Ltd.	584188.24
141	Euro+ Shopping Channel	715738.98

3. Undefined: It is the default algorithm when you create a view without specifying the Algorithm clause.It prefers Merge over Temptable because it is more efficient and it isn’t updatable if a temporary table is used.